

Proactive boundary-spanning search, organizational resilience, and radical green innovation

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Abstract

Radical green innovation has emerged as an effective strategy for firms to achieve sustainable development in the volatility, uncertainty, complexity, and ambiguity (VUCA) environment. Although the importance of external knowledge acquisition has been recognized, the mechanism of how proactive boundary-spanning search affects radical green innovation is still blurred. Drawing on the resource-based view and dynamic capabilities theory, this research discusses the impact of two types of proactive boundary-spanning search on radical green innovation and considers the mediating role of organizational resilience and the moderating role of big data analytics capability. Based on data collected from 294 manufacturing firms in China, the results show that both value-chain-based proactive search and industry-university-based proactive search positively influence radical green innovation. Furthermore, organizational resilience fully mediates the relationship between value-chain-based proactive search and radical green innovation, and partially mediates the relationship between industry-university-based proactive search and radical green innovation. Moreover, we demonstrate that big data analytics capability moderates the relationship between industry-university-based proactive search and radical green innovation. This research clarifies not only the mechanism of proactive boundary-spanning search driving radical green innovation by highlighting the mediating role of organizational resilience and the moderating role of big data analytics capability, but also provides theoretical and managerial implications for manufacturing firms and other stakeholders to carry out radical green innovation.

KEYWORDS

big data analytics capability, industry-university-based proactive search, organizational resilience, radical green innovation, value-chain-based proactive search

1 | INTRODUCTION

In order to achieve long-term and sustainable development, the top priority of firms is to carry out green innovation (Bhatia, 2021; Du et al., 2018; Huang & Li, 2017; Tang et al., 2018; Zhang & Zhu, 2019). However, the new normal of volatility, uncertainty, complexity, and ambiguity (VUCA) and the sudden outbreak of the COVID-19 crisis have brought challenges to firms in green innovation (Su et al., 2022; Zhang et al., 2022; Zhao et al., 2021), resulting in low production

efficiency (Do et al., 2022). Facing the double pressure of external environmental impact and environmental protection (Zhang et al., 2020), firms must carry out radical green innovation. Radical green innovation refers to the process in which firms develop new advanced environmental technologies by changing the original technological trajectory and fundamentally or subversively increasing the environmental benefits of existing products and processes (Dai et al., 2015). It is not only an inevitable choice for firms to break through technical bottlenecks (Su et al., 2022), but also conforms to

the innovation trend and market demand sustainability (Xin et al., 2022; Zhang et al., 2022). Therefore, how to drive firms to implement radical green innovation has emerged as a significant topic that requires to be addressed immediately.

The resource-based view suggests firms are required to pursue valuable, rare, and inimitable resources to build competitive advantage (Barney, 1991). From this perspective, existing research has analyzed the mechanism of internal heterogeneous resources on radical green innovation (Guo et al., 2020; Su et al., 2022; Zhang et al., 2022; Zhao et al., 2021; Zhou et al., 2023) and found that rich resources provide strong support for radical green innovation. Compared with incremental green innovation, radical green innovation requires more novel, complex, and extensive heterogeneous knowledge resources (Dai et al., 2015; Kiefer et al., 2019; Liao & Zhang, 2020; Martínez-Ros & Kunapatarawong, 2019). It is difficult for firms to realize radical green innovation only based on their own knowledge reserves, and firms must search for external resources across borders (Wang et al., 2020). Although some scholars found that external resource acquisition can affect radical green innovation in a beneficial way (Bhatia, 2021; Chen et al., 2014; García-Morales et al., 2012; Zhang et al., 2020), the connection between proactive boundary-spanning search and radical green innovation has received little attention. Proactive boundary-spanning search, as a new way for firms ahead of competitors to search for valuable knowledge that has not been generally recognized (O'Cass et al., 2014; Yang et al., 2021), can facilitate firms to anticipate and respond to changes in the green market environment. Moreover, firms can benefit from "first-mover advantage" by actively pursuing a proactive boundary-spanning search (Katila & Ahuja, 2002) and taking the lead in the green market (Wang et al., 2023). However, diversified and heterogeneous knowledge resources increase the search cost and innovation difficulty of firms (Katila & Ahuja, 2002; Sofka & Grimpe, 2010; Yang et al., 2021). Furthermore, scholars emphasized that knowledge acquisition from different search objects, such as value-chain partners and industry-university partners, will lead to different impacts on firm innovation (Duong et al., 2022; Köhler et al., 2012; Marchi, 2012). Specifically, knowledge acquisition from value-chain partners can help firms deeply understand green market demand and the latest trend (Ben Arfi et al., 2018; Demirel & Kesidou, 2019), and knowledge acquisition from industry-university partners can help firms quickly tackle the key core of green technologies (Ma et al., 2018). Thus, based on the search object, it is worth classifying proactive boundary-spanning search into value-chain-based proactive search and industry-university-based proactive search and exploring how two types of proactive boundary-spanning search affect radical green innovation.

Under the new normal of VUCA, firms are facing a challenging, fast-paced, and dynamic business environment (Long et al., 2023). This means that successful implementation of radical green innovation depends not only on proactive boundary-spanning search but also on firms' ability to adapt to external environments; otherwise, firms may suffer innovation crises and operational disruptions (Akgün & Keskin, 2014; Bode & Macdonald, 2017). Thus, to effectively identify and respond to external shocks, firms need to build resilience (Do

et al., 2022; Duchek, 2020). Dynamic capabilities theory points out that in a complicated and turbulent external environment, if heterogeneous resources that build competitive advantages for firms are difficult to meet the needs of the competitive environment (Eisenhardt & Martin, 2000), it will lead to core rigidity and constrain the sustainable development of firms (Song et al., 2020). Organizational resilience, as a kind of dynamic capability to withstand external environmental shocks and quickly recover from them (Williams et al., 2017), is a key source of long-term competitive advantage for firms (Teixeira & Werther, 2013). Many scholars argue that firms with high resilience are generally more proactive, flexible, and preventive (Jia et al., 2020; Parker & Ameen, 2018; Williams & Shepherd, 2016; Yao et al., 2021), as resilience can help firms timely identify emergencies and effectively respond to uncertainties and shocks in the innovation process (Do et al., 2022). This shows that firms can form organizational resilience to adapt to the external environment through proactive boundary-spanning search and thus promote radical green innovation. Even though previous research on organizational resilience has identified several antecedents, including environmental shocks (Duchek, 2020; Williams & Shepherd, 2016), organizational learning (Do et al., 2022; Linnenluecke, 2017), social capital (Jia et al., 2020; Ma & Zhang, 2022; Yu et al., 2023), and green supplier integration (Ji et al., 2020). It is still unclear how proactive boundary-spanning search drives organizational resilience. Moreover, although extensive research has emphasized that organizational resilience facilitates technological innovation (Do et al., 2022; Kantur & Say, 2015; Richtnér & Löfsten, 2014; Yang & Hsu, 2018) and environmental performance (Ji et al., 2020), it has not been adequately explained how firms can enhance organizational resilience by two types of proactive boundary-spanning search to handle the challenges associated with radical green innovation. Thus, whether and how organizational resilience mediates the impact of value-chain-based proactive search and industry-university-based proactive search on radical green innovation needs to be further explored.

In addition, with the rapid development of digital technologies, big data has become a strategic resource that nurtures green innovation (Al-Khatib, 2022; Xin et al., 2022). However, proactive boundary-spanning search is only a means of acquiring knowledge; how to effectively utilize the searched knowledge and resources mainly depends on firms' internal capabilities (Feng & Sheng, 2023; Wang et al., 2020). Dynamic capabilities theory suggests that firms' innovation depends on their capabilities to effectively manage resources (Helfat & Peteraf, 2003). As a kind of dynamic capability, big data analytics capability is becoming increasingly prominent in the digital economy (Awan et al., 2021; Gupta & George, 2016) and plays a vital role in firms' sustainable development (El-Kassar & Singh, 2019; Sahoo et al., 2023). Big data analytics capability is beneficial for firms to identify useful knowledge from voluminous data and orchestrate tangible and intangible knowledge into specific knowledge resources required for radical green innovation (Mikalef et al., 2019; Wamba et al., 2017). If firms have a high level of big data analytics capability, their efficiency in identifying and integrating external knowledge will improve (Lin & Kunnathur, 2019; Mikalef & Krogstie, 2020; Song et al., 2018)

and thus facilitate firms to conduct radical green innovation. Unfortunately, previous studies mostly focused on factors such as resource orchestration capability (Wang et al., 2020), knowledge integration capability (Sheng, 2017; Yang et al., 2021), absorptive capacity (Flor et al., 2018; Pacheco et al., 2018; Wang et al., 2022), governmental networking, and environment turbulence (Bao et al., 2021), moderated the relationship between boundary-spanning search and green innovation, and ignored the moderating role of big data analytics capability. Therefore, drawing on dynamic capabilities theory, it is necessary to address this gap by exploring the moderating effect of big data analytics capability and clarifying the impact mechanism of value-chain-based proactive search and industry-university-based proactive search on radical green innovation.

To fill the above research gaps, drawing on the resource-based view and dynamic capabilities theory, we investigate the influence of two types of proactive boundary-spanning search on radical green innovation and explore the mediating role of organizational resilience and the moderating role of big data analytics capability. This study has three main contributions as follows. First, we divide proactive boundary-spanning search into value-chain-based proactive search and industry-university-based proactive search and find that two types of proactive boundary-spanning search are important drivers of radical green innovation. Hence, our study expands the literature on proactive boundary-spanning search and provides new drivers for achieving radical green innovation. Second, there are few studies regarding organizational resilience as a mediator to link external knowledge acquisition and green innovation. Grounded on dynamic capabilities theory, we find the importance of organizational resilience by revealing the different mediating roles it plays in the relationships between value-chain-based proactive search, industry-university-based proactive search, and radical green innovation. Third, considering the digital economy, we identify the moderating role of big data analytics capability in the relationship between value-chain-based proactive search, industry-university-based proactive search, and radical green innovation, which can help us better understand the role of big data analytics capability in radical green innovation and contribute to the big data analytics capability literature.

This paper is structured as follows. First, we develop the research hypotheses. Second, we explain the methodology. Third, we provide the data analysis and results. Finally, we discuss the results and propose the research implications and limitations.

2 | RESEARCH HYPOTHESES

2.1 | Proactive boundary-spanning search and radical green innovation

According to the resource-based view, firms with the best resources and capabilities may gain comparative and competitive advantages in environmental sustainability (Grant, 1991; Serrano-García et al., 2021; Tariq et al., 2017). Compared with incremental green innovation, radical green innovation has higher complexity,

uncertainty and dual externalities (Dai et al., 2015; Martínez-Ros & Kunapatarawong, 2019), and a greater demand for heterogeneous resources. Therefore, firms are required to expand the scope of knowledge and increase the stock of heterogeneous knowledge resources (Pacheco et al., 2018; Srinivasan et al., 2021). Proactive boundary-spanning search, as the behavior of a firm to search for heterogeneous knowledge across borders ahead of its competitors (O'Cass et al., 2014), can help firms obtain knowledge and information needed for radical green innovation (Flor et al., 2018). The knowledge includes not only green needs from customers (Ben Arfi et al., 2018), environmentally-friendly raw materials from suppliers (Cao et al., 2021), and the latest trends from partners and competitors (Marchi, 2012), but also involves advanced technologies from universities and research institutes (Hu et al., 2021), the green development policies from governments (Zhang et al., 2022), and the green market prospects of management consultancies (Yu et al., 2020). Therefore, the essence of radical green innovation is a continuous process of external knowledge acquisition, and firms can acquire diverse knowledge through communication and cooperation with partners, such as customers, suppliers, other industries, universities, research institutes, governments, and management consultancies.

Existing research only explored the impact of proactive boundary-spanning search on radical green innovation from the perspective of search timing (Wang et al., 2023). However, it cannot accurately represent the influence of knowledge from different sources on radical green innovation. Firms need to deeply understand the nature and effect of knowledge search to successfully implement innovation (Katila & Ahuja, 2002; Köhler et al., 2012). Hence, the degree of heterogeneity among various knowledge sources should not be ignored. Considering search objects, we divide proactive boundary-spanning search into value-chain-based proactive search and industry-university-based proactive search. Value-chain-based proactive search is the activity by which firms ahead of competitors search knowledge from suppliers, customers, and other industries. While industry-university-based proactive search is the activity by which firms ahead of competitors search knowledge from universities, research institutes, governments, and management consultancies.

Value-chain-based proactive search is important for firms in emerging economies such as China to achieve radical green innovation, enabling firms to quickly identify innovation opportunities and commercialize new ideas (Zhang et al., 2017). It will help firms ahead of competitors to prioritize green demand and market information (Ben Arfi et al., 2018; Demirel & Kesidou, 2019) and develop new green products, thereby facilitating radical green innovation. First, value-chain-based proactive search helps firms take the lead in new markets and identify customer needs or potential market opportunities (Hoppmann et al., 2021). This search improves firms' understanding of current and potential customer needs and preferences, helps firms gain a deeper understanding of customer product usage and substitution patterns (Ciampi et al., 2021), and provides solutions for potential problems with new products (Zhang et al., 2017). Thus, by acquiring knowledge from value-chain partners, firms can develop green products to meet market needs (Sidhu et al., 2007; Wang

et al., 2023), increase the updating speed of green products (Bode & Macdonald, 2017), and then facilitate radical green innovation. Second, value-chain knowledge is usually regarded as the source of ideas for green innovation (Köhler et al., 2012; Srinivasan et al., 2021). Specifically, value-chain-based proactive search enables firms to conduct active market research or design green products to meet customer needs (Wang et al., 2019; Zhang et al., 2017), which will reduce the risk of radical green innovation failure and increase market acceptance of green products (Duong et al., 2022; Sheng, 2017), thereby achieving radical green innovation. Third, value-chain partners can continuously exchange information, develop capabilities, and learn from each other, which provides firms with better information on demand changes and new product launches (Sofka & Grimpe, 2010). In this way, by conducting value-chain-based proactive search, firms can target the correct market segments, make appropriate strategic decisions in product R&D, production, and marketing (Flor et al., 2018; Marchi, 2012), and reduce the risk of entering the green market and the risk of green innovation failure, thus enhancing radical green innovation. Based on this, we propose the following hypothesis:

H1a. Value-chain-based proactive search has a positive impact on radical green innovation.

Since the acquisition of heterogeneous resources provides more possibilities for firms to enhance green innovation (Hu et al., 2021), we think that industry–university-based proactive search may have a positive impact on radical green innovation. Specifically, on the one hand, industry–university-based proactive search can help a firm ahead of its competitors master core key technologies (Ma et al., 2018) and develop advanced green materials and cutting-edge green technologies (Duong et al., 2022). Moreover, this search can improve firms' understanding of cutting-edge frontier technology, help firms master intangible knowledge that is difficult to copy, and make use of it before competitors (Long et al., 2023). Thus, by acquiring knowledge from universities, research institutes, governments, and management consultancies, firms can identify not only information about green development policies and subsidies to discover new applications of green technology knowledge in other fields (Marchi, 2012; Wang et al., 2020; Zhang et al., 2022), but also excavate new environmental information and skills to fill their knowledge gaps and promote joint research on key core technologies (Liu, 2021; Ma et al., 2018; Yang et al., 2021), thereby promoting radical green innovation. On the other hand, knowledge from industries and universities is recognized as an important source of green technology (Köhler et al., 2012; Srinivasan et al., 2021). Industry–university knowledge can enhance not only the heterogeneous knowledge reserve of firms and upgrade and update firms' existing green innovation capabilities (Zhang & Liu, 2014), but also help firms focus their search on the most possible areas to reduce trial-and-error costs (Hu et al., 2021). In addition, industry–university knowledge enables firms to find the best resource allocation scheme, help firms break through existing technology trajectories (Duong et al., 2022; Pacheco et al., 2018), and create new green technologies, skills, and practices

(Snihur & Wiklund, 2019; Zhou et al., 2023), thus facilitating radical green innovation. Based on this, the following assumption is proposed:

H1b. Industry–university-based proactive search has a positive impact on radical green innovation.

2.2 | Proactive boundary-spanning search and organizational resilience

The word “resilience” first appeared in the fields of ecology, psychology, and engineering (Home & Orr, 1997; Meyer, 1982). With the increasing complexity of the external environment and the uncertainty of competitive markets, scholars have begun to introduce resilience into the management field (Carden et al., 2018; Dutta, 2017; Williams et al., 2017). At present, most research on organizational resilience is based on dynamic and static perspectives. Under the dynamic perspective, organizational resilience is a capability formed by the long-term, continuous, and dynamic experience of firms, which can help firms be more immune and better adjust to environmental shocks (Linnenluecke, 2017; Ortiz-de-Mandojana & Bansal, 2016). Under the static perspective, organizational resilience is a firm's characteristic that maintains active adaptability in today's challenging environment (Gao et al., 2017) and recovers, survives, and even further develops in the face of a crisis (DesJardine et al., 2019). In this study, we focus on the former perspective and consider organizational resilience as a kind of capability that can help firms be alert to changes, cope with the disturbance of uncertain events in crises, and gradually achieve self-regulation, immunity, and recovery. According to dynamic capabilities theory, in order to survive and develop, firms need to continuously acquire heterogeneous knowledge to ensure that resources can dynamically match environmental changes (Teece et al., 1997). In this way, firms can better cope with unexpected shocks in a complex and changeable environment, thereby achieving sustainable development (Eisenhardt & Martin, 2000; Helfat et al., 2009). Proactive boundary-spanning search, as a behavior of leading competitors to gain heterogeneous knowledge across organizational and technological boundaries, can promote the evolution and development of organizational resilience in uncertain environments. It provides resources for firms to dynamically match environmental changes, enabling firms to rapidly identify and respond to potential crisis events, thereby recovering from them (Ji et al., 2020; Ortiz-de-Mandojana & Bansal, 2016). Therefore, we propose that proactive boundary-spanning search is beneficial to organizational resilience.

Value-chain-based proactive search can help firms ahead of competitors better understand the changes in value-chain demand (Duchek, 2020) and choose value-chain knowledge that best matches the current external environment for decoding and absorption, which helps firms dynamically match market changes. Thus, value-chain-based proactive search can enhance organizational resilience in four aspects. First, since firms need knowledge and information to generate creative and innovative ideas to respond to potential

environmental accidents or problems (Cheng & Lu, 2017), value-chain-based proactive search can facilitate the transfer of valuable information and knowledge by communicating between firms and customers, suppliers, and other industries (Ji et al., 2020). Good interaction, communication, and cooperation make it easier for firms to identify and utilize potential resources, facilitate more effective resource allocation and knowledge transfer among partners, and form reasonable and effective risk sharing (Jia et al., 2020), thereby alleviating risks and enhancing organizational resilience. Second, value-chain-based proactive search is an advantage in resolving conflicts between firms and value-chain partners (Ben Arfi et al., 2018; Du et al., 2018), which in turn facilitates firms to improve their efficiency in dealing with environmental issues. In this way, firms can build organizational resilience. Third, value-chain-based proactive search is an interactive process, and its advantages originate from the practice of building mutual understanding and solving problems together (Jia et al., 2020). By building partnerships with other value-chain partners, firms can quickly perceive risks when crisis events occur (Bao et al., 2021), orchestrate external resources in a short time to solve emergencies (Barasa et al., 2018; Sheng, 2017), and then enhance organizational resilience (Cheng & Lu, 2017; Yu et al., 2023). Forth, value-chain-based proactive search enables firms to prepare for and adapt to crises in advance, launch various emergency plans (Bode & Macdonald, 2017), minimize the impact of the external environment, and foster organizational resilience, thereby accelerating organizational resilience. Thus, we proposed the following hypothesis:

H2a. Value-chain-based proactive search has a positive impact on organizational resilience.

Industry-university-based proactive search can help firms grasp the development trend of green technology more quickly (Hu et al., 2021) and understand the latest government policies and support information ahead of competitors, thus reducing trial-and-error costs and building organizational resilience (Zhang et al., 2022). First, in order to withstand and recover from external environmental shocks, firms need to generate radical ideas (Richtner & Löfsten, 2014). Through industry-university-based proactive search, firms can actively commercialize innovative achievements (Do et al., 2022; Wang et al., 2022), enhance product added value and competitive advantages (Flor et al., 2018; Yang et al., 2021), and then cope with turbulent market environments and promote organizational resilience (Kantur & Say, 2015). Second, industry-university-based proactive search is a learning process, and its advantages come from joint R&D and joint solutions of key core technologies (Long et al., 2023). By establishing communication and learning relationships with universities, research institutes, governments, and management consultancies, firms can acquire not only knowledge to strengthen organizational resilience and accumulate sufficient experience to cope with future emergencies (Barasa et al., 2018; Dutta, 2017), but also identify problems in time, make up for deficiencies, and quickly recover to a better state than before the crisis (Duchek, 2020),

thereby boosting organizational resilience. Based on this, the following assumption is proposed:

H2b. Industry-university-based proactive search has a positive impact on organizational resilience.

2.3 | Organizational resilience and radical green innovation

Since incremental green innovation is not enough to cope with the turbulent environment, current research believes that radical green innovation, as an innovation with both attributes of radical innovation and green innovation (Shang et al., 2020), should be taken as a key strategy for firms to realize sustainable development in the post-epidemic era (Cui & Wang, 2022). Radical green innovation technology is more difficult, more complex, and more uncertain than incremental green innovation (Liao & Zhang, 2020) and can significantly enhance the green competitive advantage of firms (Kiefer et al., 2019). Therefore, to realize radical green innovation, firms need not only to subvert the original business form and achieve higher-level innovation based on customer consumption patterns (Bhatia, 2021), but also need to consider ecological and environmental benefits (Tariq et al., 2017). Dynamic capabilities theory emphasizes that firms must deploy their resources and match their capabilities with the changing external environment to ensure competitiveness (Teece, 2018). Organizational resilience, as an important way for firms to timely respond to external environmental shocks (Duchek, 2020; Hillmann & Guenther, 2021), can help firms recover from crises and leapfrog development (Dutta, 2017). Therefore, we propose that organizational resilience can effectively facilitate radical green innovation.

First, organizational resilience can help firms capture small information and threats in complex environments (Duchek, 2020), identify potential risks in advance, take proactive early warning measures to reduce the cost of green innovation, and then shorten the R&D cycle of green products (Cheng & Lu, 2017). In this way, firms can optimize environmental management practices and mitigate the negative impact of innovation risk and crisis (Akgün & Keskin, 2014), thereby promoting radical green innovation. Second, organizational resilience can help not only firms recombine their reserves according to environmental changes and optimize production processes in real time during operations (Ji et al., 2020), but also reduce the use of natural resources and environmental pollution and timely recover renewable resources (Jia et al., 2020). As a result, firms can reduce the negative impact of their products and operations on the natural environment (Huang & Li, 2017), achieve the coordination of economic development and environmental protection, and then enhance radical green innovation. Third, organizational resilience can help firms identify potential development opportunities and new growth and quickly enter potential markets. In this way, firms can detect not only changes in customer green demand (Ben Arfi et al., 2018) and track innovation trends in their industry (Parker & Ameen, 2018), but also introduce green technologies, materials, and knowledge and upgrade and iterate

the attributes and R&D processes of existing green products (Bode & Macdonald, 2017), thus realizing radical green innovation. Based on this, we propose the following assumption:

H3. Organizational resilience has a positive impact on radical green innovation.

2.4 | The mediating role of organizational resilience

According to the resource-based view, external knowledge acquisition is the core process to help firms develop organizational resilience and innovation advantages (Wang & Ellinger, 2011). Since firms' internal resources are limited, firms must access external resources across borders to find alternatives and solutions (Yang et al., 2021; Zhang et al., 2017). As a result, firms that pay more attention to proactive boundary-spanning search can seize market demand and enhance green innovation capabilities. Under the VUCA environment, although firms can acquire heterogeneous knowledge needed for radical green innovation through proactive boundary-spanning search, heterogeneous resources are not sufficient to cope with external environmental shocks (Do et al., 2022). Firms also need to foster organizational resilience by proactive boundary-spanning search to identify and deal with dynamic and complex business environments, thus promoting radical green innovation.

Specifically, by conducting value-chain-based proactive search, firms can improve their sensitivity to changes in the demand and performance of environmental protection products in the green market (Bode & Macdonald, 2017). In this way, firms can identify and solve problems in green innovation (Cheng & Lu, 2017), such as poor communication across departments, delayed information feedback, low organizational efficiency, and decreased customer satisfaction (Marchi, 2012), thus fostering organizational resilience. Moreover, organizational resilience can help firms predict future disasters more accurately (Parker & Ameen, 2018) and buffer the negative impacts of crises through the reconfiguration of internal and external resources. Accordingly, firms can improve not only their early warning ability and help them identify customers' green needs and potential green market (Jia et al., 2020; Yao et al., 2021), but also promote them to effectively explore new solutions and timely develop new green products to meet market needs (Akgün & Keskin, 2014; Yu et al., 2023), thus enhancing radical green innovation. Therefore, we propose the following hypothesis:

H4a. Organizational resilience mediates the effect of value-chain-based proactive search on radical green innovation.

Industry-university-based proactive search is beneficial for firms to acquire industry-university knowledge ahead of competitors, and rich knowledge is conducive to firms accurately identifying the development trend of advanced green technologies (Hu et al., 2021; Ji

et al., 2020). In this way, firms can timely respond to crises and build organizational resilience (Linnenluecke, 2017). When facing emergencies, firms with organizational resilience can quickly identify potential crises, develop temporary solutions to cope with sudden crises (Kendra & Wachtendorf, 2003), and provide creative solutions for crises (Cheng & Lu, 2017). Thus, firms can timely capture and prevent not only the emergence of potential threats (Bode & Macdonald, 2017), but also break through existing technology trajectories, promote R&D of green technologies by optimizing resource reserves (Ji et al., 2020), and quickly respond to the opportunities and challenges brought by crises (Do et al., 2022; Ortiz-de-Mandojana & Bansal, 2016), thereby realizing radical green innovation. Hence, we develop the following assumption:

H4b. Organizational resilience mediates the effect of industry-university-based proactive search on radical green innovation.

2.5 | The moderating role of big data analytics capability

Big data analytics capability is the information processing capability of firms to transform resources into competitive advantages through data management, data infrastructure, and data talent (Mikalef et al., 2019; Wamba et al., 2017). The resource-based view holds that innovation is an activity that firms effectively acquire, manage, and apply heterogeneous resources to create value (Grant & Baden-Fuller, 2004). In the area of digital interconnection, big data analytics capability enables firms to use big data resources to evaluate, process, and generate value (Gupta & George, 2016; Shan et al., 2019), which can help firms enhance internal procedures, operations, and functions and better identify opportunities and apply resources (Singh & Del Giudice, 2019). Thus, we argue that firms with a high level of big data analytics capability can better capture the direction of green products, adapt more quickly to changing environments, and apply green technology knowledge and green innovation resources (Al-Khatib, 2022), thereby enhancing radical green innovation.

On the one hand, with a high level of big data analytics capability, firms can expand and deepen their relationships with customers, suppliers, and other industries. In this way, the scope of knowledge search will be expanded (Mikalef & Pateli, 2017), and the degree of resource management will be improved (Wamba et al., 2017). Thus, firms can develop, deploy, and reallocate more green innovation resources by boosting internal processes, business operations, and financial and capital effectiveness (Awan et al., 2021), and then enhance radical green innovation. Moreover, with a high level of big data analytics capability, firms can better analyze the demand direction of green products in a fast-paced business environment (Shan et al., 2019; Xin et al., 2022) and accurately predict consumers' green behavior and green consumption patterns (Sahoo et al., 2023; Xu et al., 2016), thereby boosting radical green innovation. On the contrary, with a low level of big data analytics capability, firms may find it

difficult to discover and use value-chain knowledge during innovation (Dubey et al., 2021). This may hamper firms from grasping green demands in a rapidly changing and complex environment (Al-Khatib, 2022), thereby restricting radical green innovation. Thus, when the level of big data analytics capability is high, value-chain-based proactive search has a stronger positive effect on radical green innovation. Based on this, we propose the following hypothesis:

H5a. Big data analytics capability positively moderates the relationship between value-chain-based proactive search and radical green innovation.

On the other hand, with a high level of big data analytics capability, firms can analyze and evaluate more knowledge acquired from universities, research institutes, governments, and management consultancies. Rich industry–university knowledge will help firms better cope with the risks associated with radical green innovation and increase their success. Thus, with a high level of big data analytics capability, firms can effectively identify green technology knowledge in industries, universities, and research institutions (Akhtar et al., 2018; Lin & Kunnathur, 2019) and accurately assess the feasibility and potential risks of green innovative solutions (El-Kassar & Singh, 2019; Zhang et al., 2022), so as to boost radical green innovation. Moreover, with a high level of big data analytics capability, firms can better monitor the development trend of green technology and track the use and reuse of green materials (Sahoo et al., 2023), which will help firms optimize resource usage and minimize waste, thereby enhancing radical green innovation (Riggs et al., 2023). On the contrary, with a low level of big data analytics capability, firms may be unable to identify the reliability and authenticity of external information, leading to the risk of information leakage (Papadopoulos et al., 2017). Due to the increased risks of radical green innovation, it will be difficult for firms to break the existing trajectory of green technology and engage in radical green innovation (El-Kassar & Singh, 2019; Mikalef & Krogstie, 2020). Accordingly, when the level of

big data analytics capability is high, industry–university-based proactive search has a stronger positive effect on radical green innovation. Hence, we make the following assumption:

H5b. Big data analytics capability positively moderates the relationship between industry–university-based proactive search and radical green innovation.

Based on the above analysis, the theoretical model constructed in this paper is shown in Figure 1.

3 | METHOD

3.1 | Data collection

The manufacturing industry, as an important pillar industry in China, is also the industry with serious pollution. Under the strategy of “carbon peaking and carbon neutrality,” greater demands are being put forward for the green development of manufacturing firms (Shang et al., 2020). However, faced with the impact of the environment and the dual pressures of external turbulent and environmental protection (Zhang et al., 2020), how manufacturing firms can successfully carry out radical green innovation has become an urgent problem to be solved. In order to obtain a typical sample, we selected manufacturing firms located in Guangdong, Jiangsu, Shaanxi, Yunnan, and Zhejiang. Specifically, Guangdong is located in the Pearl River Delta, and Jiangsu and Zhejiang are located in the Yangtze River Delta. These provinces reflect a high level of economic development. Shaanxi is located in the northwest of China and belongs to the Silk Road Economic Belt, with a medium level of economic development. Yunnan is located in the southwest of China and has a relatively low level of economic development. Manufacturing firms in the above provinces vary in their levels of radical green innovation, representing different levels of economic development and culture.

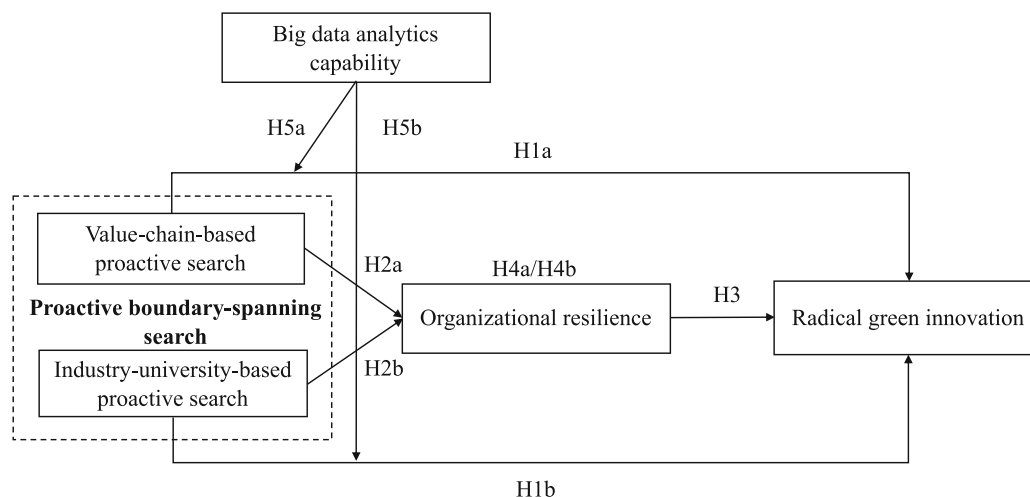


FIGURE 1 Theoretical model.

As important decision-makers in firms' development, middle and senior managers have a clear understanding of firms' current situation and development planning. Therefore, we invited middle and senior managers of manufacturing firms to fill out the questionnaire. From May to July 2022, the questionnaire was distributed in three ways. First, based on the industry directory, we randomly selected 200 firms from each region. With the help of team members, we contacted these firms to seek cooperation via e-mail or telephone. After explaining our research purpose and ensuring confidentiality, a total of 300 firms agreed to participate in the research. Then we issued the questionnaire to middle or senior managers by e-mail, and we also sent a reminder to those who did not respond 1 week later. Due to 83 firms being unable to complete the survey or having too many missing values, we ultimately collected 217 questionnaires. Second, based on the geographical advantages of our university and its affiliated research institutes, we randomly selected manufacturing firms based on the list of industrial parks in Shannxi and Jiangsu provinces. After obtaining permission from firms' executives, we interviewed the middle or senior managers of selected firms and asked them to fill out paper questionnaires on-site. We distributed 50 questionnaires, and all of them were recovered. Third, since the MBA students in our school are mainly middle and senior managers of manufacturing firms, we selected MBA students working in the above regions as the survey objects. We distributed 90 questionnaires, and 88 questionnaires were recovered. Finally, a total of 355 questionnaires were collected, and 294 valid questionnaires were completed, excluding those with regular and incomplete answers. The basic characteristics of the sample are shown in Table 1.

3.2 | Measurement

To ensure the scientificity and rationality of the questionnaire, this study mainly drew on the mature scale from existing research. The English scale was first translated into Chinese. Then, we interviewed five managers working in manufacturing firms through face-to-face communication. We asked managers to judge the rationality of the measurement items and invited managers to propose modifications to inappropriate items. Based on managers' feedback, we revised those unclear and ambiguous items, making sure that each item was understandable and relevant to the firms' actual situation. Finally, with the help of two professors who have research experience abroad, we translated the questionnaire back into English. The final questionnaire is composed of two parts: the background information of the survey firms and the content of the survey. The main variables were measured using a Likert 5 scale, and 1–5 points are assigned from *strongly disagree* to *strongly agree*.

Value-chain-based proactive search is based on the measurement scales of Katila and Ahuja (2002) and Sofka and Grimpe (2010), and four items were designed, including “Our firm can search knowledge and data about green innovation from suppliers ahead of competitors,” “Our firm can search opinions and suggestions on green innovation from customers ahead of competitors,” “Our firm can

search knowledge and data about green innovation from related industries ahead of competitors,” and “Our firm can exchange knowledge and data about green innovation from other industries ahead of competitors.” Based on the research of Katila and Ahuja (2002) and Sofka and Grimpe (2010), four items were designed for industry–university-based proactive search, including “Our firm can search knowledge and data about green innovation from universities ahead of competitors,” “Our firm can search knowledge and data about green innovation from governments ahead of competitors,” “Our firm can search knowledge and data about green innovation from research institutes ahead of competitors,” and “Our firm can search knowledge and data about green innovation from management consultancies ahead of competitors.”

Drawing on the research of Bode and Macdonald (2017) and Ji et al. (2020), four items were designed to measure organizational resilience, including “Our firm can actively pay attention to the changes of current industry situation and external environment,” “Our firm can

TABLE 1 Sample characteristics.

Characteristics of firm	Number of firms	Percentage
Firm scale		
Less than 500 employees	98	33.33%
501–1000 employees	54	18.37%
1001–2000 employees	10	3.40%
2001–3000 employees	6	2.04%
More than 3000 employees	126	42.86%
Firm age		
Less than 5 years	30	10.20%
6–10 years	48	16.33%
11–15 years	20	6.80%
16–20 years	48	16.33%
More than 20 years	148	50.34%
R&D intensity		
Less than 5%	78	26.53%
5%–10%	90	30.61%
10%–15%	22	7.48%
15%–20%	52	17.69%
More than 20%	52	17.69%
Ownership		
State-owned firms	112	38.10%
Private firms	100	34.01%
Sino-foreign firms	10	3.40%
Foreign-invested firms	72	24.49%
Region		
Guangdong	43	14.63%
Jiangsu	51	17.34%
Yunnan	61	20.75%
Shaanxi	73	24.83%
Zhejiang	66	22.45%

flexibly adapt to crisis,” “Our firm can quickly take actions to cope with crisis,” and “Our firm can easily recover operation status from crisis.”

Big data analytics capability was adapted from Akhtar et al. (2018), and eight items were designed, including “Our firm has the expertise to process structured data,” “Our firm analysts can actively gain insights from unstructured data,” “Our employees' programming skills greatly help our firm gain analytical insights from large data sets generated by intelligent devices,” “Web-based data can be effectively used by our employees to get new perspectives,” “Our firm can effectively handle complex data and information,” “Real-time information is useful for our firm's daily operations,” “Our firm's digital infrastructure attaches great importance to information integration using advanced technologies,” and “Our firm can disseminate useful information across departments.”

Four items were designed to measure radical green innovation by referring to the research of Cui and Wang (2022) and Dai et al. (2015), including “Our firm creates revolutionary new environmentally-friendly goods or services,” “Our firm launches revolutionary new concept innovations to make products or services more environmentally-friendly,” “Our firm introduces revolutionary innovations to make processes more environmentally-friendly,” and “Our firm develops and adopts revolutionary new green technologies in some fields.”

Firm age, firm size, R&D intensity, and firm ownership were selected as control variables. First, most firms lack the enthusiasm to engage in corporate social responsibility at the early stage of development (Kong et al., 2020). Thus, the longer firms are established, the more willing they are to implement radical green innovation. Second, existing research suggests that large firms have adequate financial and human resources, can access a wide range of technical opportunities and capabilities to innovate (Liao & Tsai, 2019), and are more likely to realize radical green innovation. Furthermore, radical green innovation, as an innovation process that requires multiple types of knowledge, provides good resource support for its implementation with high R&D investment (Feng et al., 2016). Therefore, R&D investment can actively facilitate firms to carry out radical green innovation and develop new green products and environmental protection processes. Finally, scholars found that state-owned firms lead the way in taking environmental responsibility in China (Bai et al., 2019) and will pay more attention to radical green innovation. We treated firm ownership as a dummy variable and analyzed state-owned firms, private firms, and foreign-invested firms by taking Sino-foreign firms as comparisons.

3.3 | Reliability and validity analysis

First, the reliability of the scale was evaluated by calculating Cronbach's alpha and composite reliability (CR), and the results are shown in Table 2. The values of Cronbach's alpha and CR are greater than 0.7 (Fornell & Larcker, 1981; Nunnally, 1978), which shows that the scale has good reliability. Second, by referring to existing

literature, expert interviews, and pre-surveys, the content validity of the questionnaire was ensured. Third, by conducting confirmatory factor analysis, the factor loading of each variable is greater than 0.5 and average variance extracted (AVE) is greater than 0.5, which demonstrated a high level of convergent validity. As for the discriminant validity, it can be seen from the results in Table 3 that the fitting values of the five-factor model are good ($\chi^2/df = 2.592$, RMSEA = 0.074, CFI = 0.913, TLI = 0.901) and significantly better than other factor models. In addition, as shown in Table 4, the square root of AVE for each variable is larger than the correlation coefficient between the corresponding variables, indicating that the scale has good discriminant validity.

3.4 | Nonresponse bias and common method

To assess nonresponse bias, early and late respondents were divided into two categories based on the return order of the questionnaire, and the *t*-test was analyzed on 50 early and 50 late respondents (Dennehy et al., 2021). The results showed that there are no significant differences in firm scale, firm age, and R&D intensity between the two groups ($p > .05$). As a result, this study has no significant non-response bias. At the same time, the study used two methods to test common method bias. First, Harman's single-factor test was used to put all the items of key variables together for factor analysis. The results showed that the maximum factor variance was 37.502%, less than 40% (Hair et al., 2016), indicating that there is no serious problem of common method bias. Second, as indicated in Table 3, confirmatory factor analysis was conducted to obtain the one-factor fitting model ($\chi^2/df = 9.494$, RMSEA = 0.170, CFI = 0.514, TLI = 0.468). Compared with other factor models, the one-factor model has the worst fit, further indicating that the problem of common method bias is not serious.

4 | RESULTS

4.1 | Descriptive statistics and correlation analysis

The mean, standard deviation, and correlation coefficient of variables are shown in Table 4. Table 4 shows that value-chain-based proactive search is positively correlated with organizational resilience ($r = .346$, $p < .01$) and radical green innovation ($r = .433$, $p < .001$). Industry-university-based proactive search is positively correlated with organizational resilience ($r = .381$, $p < .01$) and radical green innovation ($r = .508$, $p < .01$). Organizational resilience is positively correlated with radical green innovation ($r = .372$, $p < .01$).

4.2 | Hypotheses test

This study used hierarchical linear regression analysis to test hypotheses (Wang et al., 2020). As shown in Table 5, a total of

TABLE 2 Reliability and convergent validity analysis.

Variables	Items	Factor loadings	Cronbach's alpha	CR	AVE
Value-chain-based proactive search (VCPS)	VCPS1	0.719	0.746	0.811	0.518
	VCPS2	0.733			
	VCPS3	0.663			
	VCPS4	0.762			
Industry-university-based proactive search (IUPS)	IUPS1	0.738	0.838	0.831	0.556
	IUPS2	0.845			
	IUPS3	0.804			
	IUPS4	0.565			
Organizational resilience (OR)	OR1	0.807	0.869	0.870	0.626
	OR2	0.771			
	OR3	0.784			
	OR4	0.803			
Big data analytics capability (BDA)	BDA1	0.801	0.917	0.904	0.540
	BDA2	0.681			
	BDA3	0.725			
	BDA4	0.703			
	BDA5	0.756			
	BDA6	0.766			
	BDA7	0.690			
	BDA8	0.750			
Radical green innovation (RGI)	RGI1	0.635	0.826	0.829	0.552
	RGI2	0.658			
	RGI3	0.840			
	RGI4	0.817			

TABLE 3 Discriminant validity analysis.

Model	χ^2	df	RMSEA	TLI	CFI
One-factor model (VCPS + IUPS + OR + BDA + RGI)	2392.578	252	0.170	0.468	0.514
Two-factor model (VCPS + IUPS, OR + BDA + RGI)	1983.127	251	0.153	0.568	0.607
Three-factor model (VCPS + IUPS, OR, BDA + RGI)	1661.882	249	0.139	0.645	0.679
Four-factor model (VCPS + IUPS, OR, BDA, RGI)	1241.807	246	0.118	0.746	0.774
Five-factor model (VCPS, IUPS, OR, BDA, RGI)	627.347	242	0.074	0.901	0.913

Abbreviations: BDA, big data analytics capability; IUPS, industry-university-based proactive search; OR, organizational resilience; RGI, radical green innovation; VCPS, value-chain-based proactive search.

eight models were designed in the regression; the dependent variables of Models 1 and 2 were organizational resilience, and the dependent variables of Models 3–8 were radical green innovation. In Models 1 and 3, only control variables were included; organizational resilience and radical green innovation were set as the dependent variables, respectively. In Models 2 and 4, we put value-chain-based proactive search and industry-university-based proactive search into the regression as independent variables. Organizational resilience was entered in Models 5 and 6 to test the mediating of organizational resilience. To test the moderating effect, big data analytics capability was added in Model 7, and two

interaction terms were added in Model 8. The regression results are shown in Table 5.

Hypotheses [H1a](#) and [H1b](#) predicted that value-chain-based proactive search and industry-university-based proactive search have positive influences on radical green innovation. Model 4 shows that both value-chain-based proactive search and industry-university-based proactive search have positive influences on radical green innovation ($\beta = .139, p < .01$; $\beta = .440, p < .001$). Moreover, industry-university-based proactive search has a greater impact on radical green innovation, proving [H1a](#) and [H1b](#). Hypotheses [H2a](#) and [H2b](#) predicted that value-chain-based proactive search and industry-

TABLE 4 Means, standard deviations, and correlations.

Variables	1	2	3	4	5	6	7	8	9	10	11
1. Firm scale	1										
2. Firm age	0.564**	1									
3. R&D intensity	0.293**	0.112	1								
4. State-owned firms	0.441**	0.576**	0.085	1							
5. Private firms	0.039	0.026	−0.012	−0.135*	1						
6. Foreign-invested firms	−0.105	0.252**	−0.183**	−0.409**	−0.107	1					
7. VCPS	0.109	0.048	0.143**	0.118*	0.102	−0.004	0.720				
8. IUPS	0.214**	0.144*	0.322**	0.029	−0.074	−0.138*	0.602**	0.746			
9. OR	0.116*	0.189**	0.066	−0.118**	0.062	0.145*	0.346**	0.381**	0.791		
10. BDA	0.408**	0.302**	0.289**	−0.155**	−0.054	−0.002	0.449**	0.589**	0.518**	0.735	
11. RGI	0.080	0.039	0.052	0.147*	−0.166	0.012	0.433**	0.508**	0.372**	0.533**	0.743
Mean	3.027	3.803	2.694	0.340	0.034	0.245	3.345	2.968	3.889	3.573	3.582
S.D.	1.806	1.453	1.472	0.475	0.182	0.431	0.729	0.811	0.789	0.863	0.773

Note: The diagonal numbers in bold are the square root of AVE.

Abbreviations: BDA, big data analytics capability; IUPS, industry–university-based proactive search; OR, organizational resilience; RGI, radical green innovation; VCPS, value-chain-based proactive search.

* $p < .05$, and ** $p < .01$.

TABLE 5 Regression results.

Variables	OR		RGI					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Control variables								
Firm scale	0.062	−0.003	0.179*	0.106	0.155*	0.106	0.001	−0.018
Firm age	0.149	0.087	0.085	0.006	0.029	−0.012	−0.020	0.010
R&D intensity	0.070	−0.037	0.040	−0.099	0.013	−0.091	−0.125*	−0.142**
State-owned firms	0.082	−0.020	0.318***	0.206**	0.286***	0.210**	0.206**	0.251**
Private firms	0.086	0.115*	−0.117*	−0.082	−0.150**	−0.105*	−0.078	−0.082
Foreign-invested firms	0.170*	0.163*	0.133	0.140*	0.069	0.107	0.108	0.099
Independent variables								
VCPS		0.178**		0.139**		0.102	0.075	0.085
IUPS		0.305***		0.440***		0.378***	0.266***	0.258***
Mediator								
OR					0.381***	0.204***		
Moderator								
BDA							0.413***	0.442***
Interactions								
VCPS × BDA								0.107
IUPS × BDA								0.153**
R^2	0.060	0.225	0.088	0.333	0.224	0.365	0.425	0.434
ΔR^2	0.060	0.165	0.088	0.245	0.205	0.141	0.407	0.009
F	3.072**	10.345***	4.589***	17.758***	11.768***	18.139***	23.325***	19.650***

Abbreviations: BDA, big data analytics capability; IUPS, industry–university-based proactive search; OR, organizational resilience; RGI, radical green innovation; VCPS, value-chain-based proactive search.

* $p < .05$, ** $p < .01$, and *** $p < .001$.

university-based proactive search have positive effects on organizational resilience. As shown in Model 2, both value-chain-based proactive search and industry-university-based proactive search have positive effects on organizational resilience ($\beta = .178, p < .01$; $\beta = .305, p < .001$), thus supporting H2a and H2b. Hypothesis H3 argues for the positive impact of organizational resilience on radical green innovation. As Model 5 displayed in Table 5, organizational resilience has a significant positive impact on radical green innovation ($\beta = .381, p < .001$), and H3 is supported.

Hypotheses H4a and H4b predicted that organizational resilience mediates the effects of value-chain-based proactive search and industry-university-based proactive search on radical green innovation. According to Baron and Kenny's (1986) method, we used hierarchical regression to examine mediating hypotheses. First, we set radical green innovation and organizational resilience as dependent variables, then added the independent variables of value-chain-based proactive search and industry-university-based proactive search, respectively, and finally put the mediating variable of organizational resilience into regression. As mentioned above, Model 4 shows that value-chain-based proactive search and industry-university-based proactive search have significant positive impacts on radical green innovation ($\beta = .139, p < .01$; $\beta = .440, p < .001$). Model 5 shows that organizational resilience has a significant positive effect on radical

green innovation ($\beta = .381, p < .001$). As can be seen from Model 6, when organizational resilience was added, industry-university-based proactive search still has a significant positive effect on radical green innovation ($\beta = .378, p < .001$). However, the effect of value-chain-based proactive search on radical green innovation was not significant ($\beta = .102, p > .05$) after adding organizational resilience. Thus, organizational resilience plays a fully mediating role between value-chain-based proactive search and radical green innovation, and a partial mediating role between industry-university-based proactive search and radical green innovation. Therefore, H4a and H4b are supported.

Furthermore, we used the Bootstrap method with 5000 replications and a 95% confidence interval (CI) to further test the mediating effect of organizational resilience between value-chain-based proactive search, industry-university-based proactive search, and radical green innovation (Hayes & Preacher, 2010). As displayed in Table 6, the indirect effect coefficient of value-chain-based proactive search is .0594, and the CI is [.0151, .1162], excluding 0. The indirect effect coefficient of industry-university-based proactive search is .0386, the CI is [.0045, .0849], and 0 is excluded, supporting H4a and H4b.

Hypotheses H5a and H5b posited that big data analytics capability moderates the relationship between value-chain-based proactive search and radical green innovation, as well as industry-university-

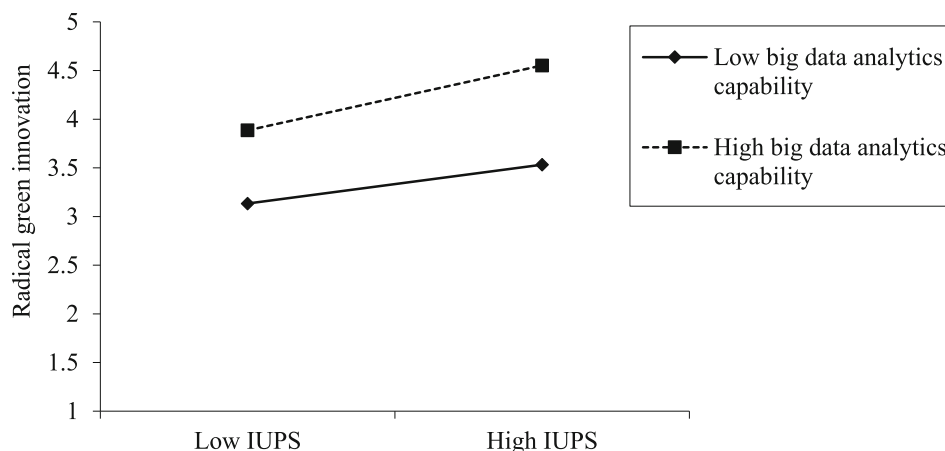
TABLE 6 Bootstrap results of mediating effects.

Path	Direct effect	SE	t	LICL	ULCI
IUPS → RGI	0.3600	0.0624	5.7687***	0.2372	0.4828
VCPS → RGI	0.1082	0.0654	1.6546	−0.0205	0.2369
	Indirect effect	Boot SE		Boot LCLC	Boot UCLC
IUPS → OR → RGI	0.0594	0.0259		0.0151	0.1162
VCPS → OR → RGI	0.0386	0.0205		0.0045	0.0849
	Total effect	SE	t	LCLC	UCLC
IUPS → RGI	0.4194	0.0618	6.7838***	0.2977	0.5411
VCPS → RGI	0.1468	0.0661	2.2211*	0.0167	0.2770

Abbreviations: IUPS, industry-university-based proactive search; OR, organizational resilience; RGI, radical green innovation; VCPS, value-chain-based proactive search.

* $p < .05$, and *** $p < .001$.

FIGURE 2 The moderating effect of big data analytics capability between industry-university-based proactive search and radical green innovation.



based proactive search and radical green innovation. First, radical green innovation was added to the regression model as a dependent variable, and independent variables value-chain-based proactive search and industry-university-based proactive search were gradually added, and then the moderating variable of big data analytics capability was entered. Finally, the interactions of value-chain-based proactive search and big data analytics capability, as well as industry-university-based proactive search and big data analytics capability, were put into the regression model. Model 8 shows that the interaction between value-chain-based proactive search and big data analytics capability has no significant impact on radical green innovation ($\beta = .107, p > .05$), thus indicating big data analytics capability has no significant impact on the relationship between value-chain-based proactive search and radical green innovation. The interaction between industry-university-based proactive search and big data analytics capability has a significant positive impact on radical green innovation ($\beta = .153, p < .01$), indicating big data analytics capability positively moderates the relationship between industry-university-based proactive search and radical green innovation. Thus, H5b is supported, whereas H5a is not supported. To further explain the moderating effect, we used the method of Dawson and Richter (2006) to plot the effects of different levels of big data analytics capability on the relationship between industry-university-based proactive search and radical green innovation. As shown in Figure 2, compared with low big data analytics capability, the positive relationship between industry-university-based proactive search and radical green innovation is more significant when the level of big data analytics capability is high.

5 | DISCUSSION, CONCLUSION, AND LIMITATION

5.1 | Discussion

Combining resource-based view and dynamic capabilities theory, this study explores the relationship between two types of proactive boundary-spanning search and radical green innovation, and investigates the mediating effect of organizational resilience and the moderating effect of big data analytics capability.

First, the results show that both value-chain-based proactive search and industry-university-based proactive search have positive effects not only on radical green innovation but also on organizational resilience. The findings support the conclusion of Ben Arfi et al. (2018), who argued that different types of external knowledge, such as knowledge gained from universities, research institutions, firms, and consumers, have different effects on green innovation. Our results are also partially consistent with the research of Zhang et al. (2017), who confirmed that boundary-spanning demand-side search guides firms to develop advanced technologies to meet green needs, and expand the research of Yang et al. (2021), who suggested that moderate proactive boundary-spanning search can help firms develop green innovation. According to the search object, we further divide proactive boundary-spanning search into value-chain-based proactive

search and industry-university-based proactive search. This is because, as the diversity of externally available knowledge increases, it becomes more difficult for firms to search for and integrate knowledge (Köhler et al., 2012; Marchi, 2012). Boundary-spanning search based on different search objects is conducive to improving the search efficiency of firms and reducing search costs and innovation risk (Duong et al., 2022; Zhang et al., 2017). We find that it is beneficial for firms to dynamically match the changes of the external environment and cultivate organizational resilience through proactive boundary-spanning search. This is consistent with Duchek (2020), who suggested that internal and external knowledge resources are particularly significant for building organizational resilience. Our finding also extends the study of Cheng and Lu (2017), who proposed that the ability of firms to actively acquire knowledge from partners can ensure the survival and development of firms, thus forming organizational resilience.

Second, our results indicate that organizational resilience positively influences radical green innovation. Specifically, organizational resilience can guide firms to identify, withstand, and recover from innovation crises by dynamically matching heterogeneous resources with a competitive environment. Previous research studies on organizational resilience and firms' innovation mainly regarded innovation as a single dimension, which may obscure the impact of organizational resilience on different innovations, especially radical green innovation. We prove that organizational resilience is crucial for firms to realize radical green innovation. Therefore, our result supports the argument of Kantur and Say (2015) and Richtnéř and Löfsten (2014), who believed that organizational resilience plays an important role in coping with innovation risks and improving firms' creativity. Taking radical green innovation into account, our finding is partially consistent with the conclusion of Ji et al. (2020), who emphasized that organizational resilience can promote the sustainable development of firms.

Third, our results reveal that organizational resilience mediates the relationship between value-chain-based proactive search, industry-university-based proactive search, and radical green innovation. Specifically, organizational resilience fully mediates not only the relationship between value-chain-based proactive search and radical green innovation, but also partially mediates the relationship between industry-university-based proactive search and radical green innovation. This is consistent with the research of Do et al. (2022), who suggested that organizational resilience plays a mediating role between organizational learning and innovation. We find that value-chain-based proactive search has an indirect effect on radical green innovation, and organizational resilience plays a fully mediating role. The possible reason is that green market demand changes faster in the VUCA environment, and knowledge obtained by value-chain-based proactive search, such as green group thinking, expectations, consumption preferences, and competitors' conditions, is more susceptible to the external environment. Hence, value-chain-based proactive search affects organizational resilience, and organizational resilience further affects radical green innovation. Our research strengthens the opinion that external knowledge acquisition is conducive to building organizational resilience and explains how

firms can leverage heterogeneous knowledge to facilitate radical green innovation.

Fourth, our results show that big data analytics capability plays a moderating role in the relationship between industry–university-based proactive search and radical green innovation. The finding supports the results of Al-Khatib (2022), Mikalef et al. (2019), and Wamba et al. (2017), who proposed that big data analytics capability can accelerate the transformation of existing knowledge into firm performance and competitive advantage. The finding is also consistent with Xin et al. (2022), who argued that firms can acquire complementary green resources and knowledge more effectively with a high level of big data analytics capability. This is because most industry–university knowledge obtained from universities, research institutes, governments, and management consultancies is theoretical and standardized (Köhler et al., 2012). In other words, industry–university knowledge has greater technical difficulty and a longer technological cycle. If firms want to apply this kind of knowledge frequently, they must rely on big data analytics capability. With a high level of big data analytics capability, firms can effectively transform and apply industry–university knowledge to better carry out radical green innovation. While we find that big data analytics capability does not moderate the link between value-chain-based proactive search and radical green innovation, which is partially opposite with Ciampi et al. (2021), who found that big data analytics capability can strengthen the relationship between firms and other stakeholders and help firms create better value. One possible explanation is that most knowledge acquired from customers, suppliers, and other industry firms is value-chain-based knowledge, which has a greater impact on maintaining sustainable competitive advantage. This also explains the research of Ciampi et al. (2021), who argued that big data analytics capability spurs innovation to enhance firms' competitive advantage. Moreover, the synergy between knowledge provided by value-chain partners and the existing knowledge base is the strongest (Ben Arfi et al., 2018). Thus, knowledge from value-chain partners is more actionable, which makes it easier for firms to commercialize. As a result, value-chain-based proactive search can drive radical green innovation regardless of the level of big data analytics capability. Therefore, big data analytics capability has no significant moderating effect on the relationship between value-chain-based proactive search and radical green innovation.

5.2 | Theoretical implications

This study contributes to the literature in several ways. First, prior studies on proactive boundary-spanning search mainly focused on search timing (Wang et al., 2023; Yang et al., 2021) and paid less attention to search objects. Although scholars suggest that where to search for valuable knowledge is crucial for firms to innovate (Duong et al., 2022; Köhler et al., 2012; Zhang et al., 2017), the impact of different search objects on radical green innovation is still unclear. By considering search objects, we distinguish two types of proactive boundary-spanning search, namely, value-chain-based

proactive search and industry–university-based proactive search, to deeply understand proactive boundary-spanning search. Thus, this study enriches the dimensions of proactive boundary-spanning search. Furthermore, despite the fact that existing studies have concentrated on the factors driving green innovation (Bhatia, 2021; Zhang et al., 2020), the literature on radical green innovation with high technology is still limited. From search object and proactive behavior perspectives, this study regards value-chain-based proactive search and industry–university-based proactive search as key drivers of radical green innovation. Hence, these findings further supplement not only the research on proactive boundary-spanning search by considering search objects, but also offer a novel viewpoint to comprehend the drivers of radical green innovation.

Second, our results unfold the “black box” in the relationship between value-chain-based proactive search, industry–university-based proactive search, and radical green innovation by investigating the mediating role of organizational resilience. Although existing research has focused on the direct effect of external knowledge acquisition on radical green innovation (Chen et al., 2014; Zhang et al., 2020), it is still unclear what mechanism and how value-chain-based proactive search and industry–university-based proactive search affect radical green innovation. Due to the uncertainty of the external environment, it is increasingly common for firms to face shocks (Linnenluecke, 2017; Williams & Shepherd, 2016). Therefore, this study emphasizes the necessity of organizational resilience. Based on dynamic capabilities theory, we explore how organizational resilience affects the relationship between value-chain-based proactive search, industry–university-based proactive search, and radical green innovation. The results show that organizational resilience plays a fully mediating role between value-chain-based proactive search and radical green innovation, and plays a partial mediating role between industry–university-based proactive search and radical green innovation. This study elucidates not only the role of organizational resilience, but also provides new insights for us to understand how different types of proactive boundary-spanning search affect radical green innovation.

Third, this study highlights the role of big data analytics capability and uncovers that the relationship between industry–university-based proactive search and radical green innovation is moderated by big data analytics capability. Although some scholars confirmed that big data analytics capability contributes to resource acquisition and integration (Al-Khatib, 2022; Gupta & George, 2016), there is little research on how big data analytics capability can moderate the relationship between value-chain-based proactive search, industry–university-based proactive search, and radical green innovation. Drawing on dynamic capabilities theory, we find that big data analytics capability positively moderates the relationship between industry–university-based proactive search and radical green innovation. This result expands not only the moderating factors that influence the relationship between industry–university-based proactive search and radical green innovation, but also uncovers how big data analysis capability can strengthen the impact of industry–university-based

proactive search on radical green innovation and expands the application field of dynamic capabilities theory.

5.3 | Managerial implications

Our results provide some managerial implications for manufacturing firms, universities, and policymakers. First, we find that organizational resilience can help firms actively evaluate and digest heterogeneous knowledge and dynamically match changes in the external environment, so that firms can successfully carry out radical green innovation. Hence, in order to survive in the volatile environment, especially under the new normal of VUCA, manufacturing firms need to strengthen their early warning and response capabilities for crisis events. Specifically, firms need not only to smooth information exchange channels and establish a real-time warning system to help firms monitor early crisis signals and timely detect abnormal events, but also to strengthen employees' crisis awareness and improve employees' emergency management capabilities. In this way, firms can quickly develop detailed emergency plans in the face of emergencies. Moreover, we find that with the help of big data analytics capability, manufacturing firms can efficiently integrate and apply resources and improve radical green innovation. Therefore, we suggest manufacturing firms should pay close attention to the development of big data technology and the cultivation of big data talents. Specifically, firms need not only to strengthen R&D intensity in the field of big data and actively develop or purchase big data technology products, but also need to increase training in business intelligence, data mining, and basic statistics and recruit talents with good technical and management skills in big data analytics, hence optimizing firms' ability to use big data to support decision-making.

Second, we find that proactive boundary-spanning search is a critical driver for promoting radical green innovation. It is suggested that effective external knowledge acquisition can alleviate resource bottlenecks in the innovation process, which is vital for manufacturing firms to realize radical green innovation. Therefore, as an important objective of proactive boundary-spanning search, universities should actively participate in green cooperation activities between manufacturing firms and other stakeholders. Specifically, on the one hand, universities can actively develop green invention patents through funds established by firms. This can guide universities not only to participate in green innovation through industry-university cooperation, but also to introduce and cultivate high-tech talents for firms through joint training. On the other hand, universities can regularly hold academic seminars to promote resource integration and information sharing and facilitate radical green innovation. In this way, universities can grasp the development trend of green technology through in-depth communication with other stakeholders, thereby accelerating joint research on key technologies and the transformation of scientific research achievements.

Third, our research also has practical implications for policymakers. Although the cost and risk of radical green innovation are relatively high, it can bring greater technological leaps and potential

social benefits to firms. Thus, as the maker of laws, regulations, and policies, the government should increase its support for green innovation and actively create a better policy environment for sustainable development. The government can strengthen financial support for green development and encourage firms to invest more resources in radical green innovation through tax deductions and exemptions and government subsidies. Meanwhile, to better encourage radical green innovation, the government should improve the market mechanism, optimize the technology exchange environment, and strengthen the protection of green intellectual property by enforcing stricter regulations. In this way, managers will increase investment confidence and guide manufacturing firms to carry out radical green innovation.

5.4 | Limitations and future research

The limitations of this study offer a number of options for further research. First, we selected manufacturing firms as research objects, which is an important pillar industry in China. Under the strategy of "carbon peaking and carbon neutrality goals," the sustainable development of manufacturing firms is supervised and supported by various ministries and commissions. Thus, our research results may have certain industry limitations, and the generalization needs to be further tested. Future research should collect relevant data from different countries or industries to improve the generalizability of the research. Meanwhile, although our research took firm size as a control variable, we found that it has no impact on radical green innovation. Therefore, future research can explore this relationship in other industries. Besides, our research does not consider the business model, and future research should discuss the differences between various business models in terms of radical green innovation.

Second, this research only reveals the mediating role of organizational resilience and the moderating role of big data analytics capability. Since radical green innovation is a result of many factors (Su et al., 2022), it is necessary to seek other potential variables that may influence the relationship between value-chain-based proactive search, industry-university-based proactive search, and radical green innovation. Some scholars emphasized that effective resource management also contributes to sustainable development (Cui & Wang, 2022; Long et al., 2023). Thus, future research can consider the mediating role of green knowledge integration and environmental resource orchestration. Meanwhile, due to the complex and changeable external environment, the impact of external environmental variables on radical green innovation should not be ignored. Future research can explore the moderating role of environmental competitiveness, market attractiveness, and the institutional environment, which can help us better understand the benefits and risks of radical green innovation.

Third, proactive boundary-spanning search is not a perfect search, and excessive search may lead to the risk of distraction. Specifically, in order to occupy the green market, firms may spend more time, energy, and financial resources to acquire and learn heterogeneous knowledge. This will distract managers' attention, causing firms to be

unable to timely match new green knowledge with changes in the dynamic environment, thereby hindering the sustainable development of firms to a certain extent (Yang et al., 2021). Therefore, future research should consider the possible negative effects of two types of proactive boundary-spanning search and explore how firms can avoid such risks in radical green innovation.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there are no conflicts of interest.

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